

Kinetic Parameter Estimation of the Polyethylene Process by Bayesian Optimization

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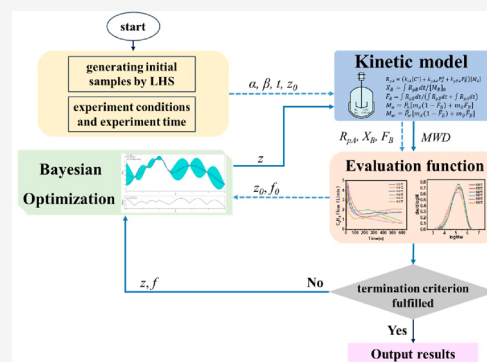


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ABSTRACT: The design and optimization of polymerization processes rely on accurate kinetic parameters of polymerization reactions since they have a significant impact on polymer properties. We develop a kinetic parameter estimation methodology for non-steady-state olefin polymerization processes. A dynamic model for olefin polymerization reaction kinetics based on the method of moments and instantaneous distribution method is established, which considers reaction temperature fluctuations and concentration variation during polymerization. Bayesian optimization (BO) algorithm is applied in the parameter estimation framework, where the kinetic model is treated as a black box function. The effectiveness of the method developed in this article was demonstrated through three cases, and the parameter estimation strategies used in other literature studies were compared. Additionally, we conduct a comparison of the fitting and extrapolation results obtained from diverse algorithms. Notably, BO emerges as a favorable alternative to deterministic algorithms, as it circumvents the challenges inherent in utilizing gradient-based optimization methods for complex polymerization dynamics models and is more suitable for scenarios with fast prediction in high-throughput experiments. Furthermore, using a dynamic kinetic model is crucial for kinetic parameter estimation in non-steady-state olefin polymerization, as it can reflect the dynamic changes with reaction temperature and concentrations.



1. INTRODUCTION

The mechanism of olefin polymerization is complex, and the key properties of the polymer cannot be measured in a timely manner, such as degree of polymerization, molecular weight distribution (MWD), etc. Consequently, the design and optimization of the polymerization process heavily depend on mathematical models and process optimization. Establishing an accurate polymerization kinetic model holds significant importance, as the polymerization kinetic model determines the microscopic properties of the polymer product, such as the melt index (MI) and polydispersity index (PDI). Microscopic properties consequently affect the macroscopic properties of the product, such as tensile strength and yield strength. Obtaining precise kinetic parameters is of great significance for the application of the model. However, the intricate nature of polymerization reactions necessitates the prediction of many kinetic parameters. Obtaining a comprehensive set of kinetic parameters through a conventional method of fitting the kinetic curves is impractical. Several studies have explored kinetic parameter estimation in olefin polymerization, employing both self-developed models and commercial software (e.g., Aspen Polymer).

Sharma¹ utilized simulation tools, sensitivity analysis, design specifications, and data fitting functions within Aspen Polymer to demonstrate a computer-aided parameter estimation

method, which considers commercially significant production targets. It involves parameter estimation for both single-active site and multi-active site catalysts, applied in industrial processes for HDPE, PP, and LLDPE production. While commercial process software resolves certain issues, it falls short in customizing objective functions for optimization, such as objective functions considering MWD.

When estimating kinetic parameters, it necessitates deconvolution of the gel permeation chromatography (GPC) curve² to ascertain the number of active sites. The MWD of the polymer generated by a single-site catalyst conforms to the Schulz–Flory distribution.^{3,4} Several methodologies for the kinetic parameter estimation and deconvolution have been proposed. Touloupidis⁵ introduced an effective methodology capable of estimating all pertinent parameters and reverse engineer experimental results. Feng^{6,7} carried out parameter estimation centered on a non-standard ethylene/α-olefin gas phase copolymerization reaction mechanism, utilizing MWD

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and short chain branch distribution (SCBD) as objective functions. However, relying on MWD or microscopic chain structure information as the sole evaluation function for optimization might be insufficient. The obtained kinetic parameter lacks the necessary accuracy to predict the consumption of monomers and comonomers. Liu⁸ and Luo⁹ utilize the method of moments and population balance model to establish the kinetic models of ethylene/octene copolymerization and tandem polymerization of ethylene, respectively. They perform parameter estimation by fitting experimental data and establishing a precise polymerization reaction model for guiding feeding strategy in experimental process aimed at producing specific products. Gibson^{10,11} developed kinetic models for gas-phase ethylene/1-hexene copolymerization employing a 3-site hafnocene catalyst. Taking the deviation of MWD and comonomer composition as the objective function. The study prioritized parameters by importance and proposed an effective method that combined dynamic and steady-state data, with the aim of crafting a more accurate mathematical model.

When parameter estimation is performed, the fitted kinetic parameters typically encompass the pre-exponential factors of the Arrhenius equation, while the apparent parameters are computed through equations. The nature of the Arrhenius equation reveals that the apparent parameters are affected by reaction temperature, thereby impacting both the rate of polymerization reaction and product properties.^{12–14} Previous studies have commonly assumed a constant reaction temperature in their work.¹⁵ Caldera¹⁶ fitted the ethylene consumption curve during the ethylene-hexene solution polymerization reaction across temperatures of 100 °C, 120 °C, and 140 °C. They obtain the pre-exponential factors of chain propagation parameters and chain deactivation parameters and the corresponding activation energy values. Wu¹⁷ investigated the ethylene-hexene polymerization process, employing an identical kinetic expression to fit the polymerization rate curves at two distinct temperatures. However, some polymerization systems experience temperature variations throughout the polymerization reaction, which has an impact on the polymerization reaction kinetics. This phenomenon has frequently been overlooked in prior studies.

When the kinetic parameters of the polymerization reaction are estimated, numerous kinetic parameters and activation energy parameters are involved. The vast parameter space presents challenges in achieving optimal solutions and demands an extensive computational time. Hence, the selection of an appropriate optimization algorithm and strategy becomes pivotal in ensuring accurate estimations. Due to the highly nonconvex and nonlinear nature of the problem, employing gradient-based optimization algorithms for parameter estimation encounters significant challenges. And gradient-based optimization algorithms generally rely on initial points and require relatively more time than derivative-free optimization. Although the importance of accuracy in kinetic parameter estimation outweighs the computational time in some scenarios, the computational time is equally significant in scenarios of automated high-throughput experiments. For example, kinetic parameters can be estimated using multiple parallel experiments, which will increase the number of variables and constraints and pose challenges for gradient-based optimization algorithms. Developing gradient-free optimization based kinetic parameter estimation methods can balance solving speed and accuracy, which is of great

significance for its application. Bayesian optimization (BO),^{18,19} one of the versatile derivative-free optimization algorithms, is a highly promising global optimization method. It has been widely applied in artificial intelligence, machine learning, and chemical process optimization. Urm²⁰ employed BO method to optimize the chemical recycling process of polyethylene terephthalate (PET), aiming to minimize the total annualized cost. Ward²¹ utilized BO algorithm to minimize the CO₂ capture cost of a post-combustion carbon capture system. The application of the BO algorithm in kinetic parameter estimation has rarely been reported, especially for polyethylene reactions. The vast kinetic parameter space and multiple objective functions pose challenges in utilizing BO for the kinetic parameter estimation of polyethylene processes. A prevalent strategy involves ranking kinetic parameters according to their impact on the target value,^{6,11} followed by a stepwise process of fixing parameters for fitting. Yet, determining hyperparameters for weight coefficients within a singular objective function and ranking kinetic parameters prove challenging. Moreover, previous kinetic models mostly neglect the impact of reaction temperature fluctuations and concentration variations during polymerization. This oversight within existing parameter estimation strategies exhibits issues such as poor extrapolation, leaving significant room for optimization.

In light of the aforementioned concerns, a dynamic model of olefin polymerization reaction kinetics is developed by employing mathematical modeling tools, which considers the impact of temperature changes on kinetic behavior. A kinetic parameter estimation framework based on the BO algorithm is constructed where step-by-step estimation of different kinetic parameter subsets is performed using multiple production indicators as evaluation criteria for each subset. Both the pre-exponential factor and the activation energy are estimated. Three case studies are carried out to demonstrate the effectiveness of the proposed method. Comparative analyses against other estimation strategies and algorithms show distinct advantages of the proposed method, particularly in its better extrapolation performance.

2. PARAMETER ESTIMATION FRAMEWORK

A typical parameter estimation problem can be defined as below:

$$\begin{aligned} \min f(R_{pA}, X_B, F_B, \text{MWD}, \beta) \\ \text{s. t.} \quad R_{pA} = g(z, t, \alpha) \\ X_B = h(z, t, \alpha) \\ F_B = n(z, t, \alpha) \\ \text{MWD} = m(z, t, \alpha) \end{aligned} \quad (1)$$

where the function f is the evaluation function, which can be calculated from the process variables R_{pA} , X_B , F_B , and MWD and the known experiment results β . The evaluation function reflects the accuracy of the simulation results. R_{pA} , X_B , F_B , and MWD are the monomer reaction rate (R_{pA} , time-dependent), comonomer conversion rate (X_B), polymer comonomer content (F_B) and the production MWD, calculated by the polymerization reaction kinetic model of nonlinear functions g , h , n , and m . Parameters α and t represent the experimental conditions and the experiment time, respectively. Note that experiment conditions can be time-dependent, such as reaction

temperatures. Decision variables z are kinetic parameters to be estimated, including the pre-exponential factor k_0 and the activation energy E_a of each polymerization reaction. The apparent rate constant k of each reaction can be obtained by the well-known Arrhenius equation, as shown by eq S1 in the Supporting Information.

For semibatch olefin polymerization experiments, the polymerization reaction kinetic model is a dynamic model that includes differential algebraic equations (DAEs). Discretizing the DAEs of the kinetic model results in a large-scale nonlinear programming problem. When the kinetic parameter estimation method is deployed in an automated high-throughput experiment, it is generally utilized for estimating kinetic parameters through multiple parallel experiments, where the number of variables and constraints increases with the number of experiments. Solving such complex optimization problems using deterministic algorithms is computationally intensive and challenging. To overcome the challenge, a simulation-based optimization method for the polymerization kinetic estimation via BO is developed, as shown in Figure 1.

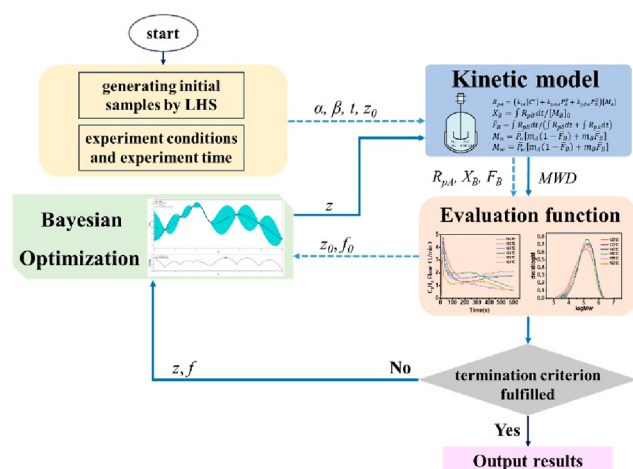


Figure 1. Parameter estimation framework.

The experimental conditions α and experiment time t will be sent to the dynamic polymerization model. The first step of the optimization involves generating the initial samples for the BO, indicated by the dashed blue line. Initial decision variables z_0 are generated by sampling methods such as Latin Hypercube Sampling (LHS). The process variables are derived from the dynamic polymerization model, enabling calculation of the evaluation function f . The initial samples of (z_0, f_0) are used to initialize the BO posterior distribution. The second step comprises the iterative update of new decision variable z , simulation of DAEs, and calculation of evaluation function f , which are represented by the solid blue line. The iteration will stop if the evaluation function meets the termination criterion or if the iteration count exceeds the specified maximum limit.

2.1. Polymerization Reaction Kinetic Model. In the parameter estimation framework, the dynamic polymerization model is regarded as a black box model for optimization. The model is established based on the coordination polymerization mechanism of metallocene and Ziegler–Natta catalysts. To simplify the model, the following assumptions^{1,8,16,22} are considered:

1. The metallocene catalyst is activated instantaneously, that is, the chain activation rate constant is relatively

large. No chain initiation occurs at the comonomer end groups.

2. The thermal deactivation of the catalyst is not considered.
3. The reactor is considered a fully mixed reactor, where the concentration and temperature are the same in the entire reactor.
4. Mass transfer resistance is disregarded within the system. The gas and liquid phases are consistently in equilibrium, implying that the amount of ethylene flowing into the reactor is regarded as the consumption.
5. The chain propagation reaction adopts the terminal model, that is, the copolymerization kinetics depend on the type of comonomer at the growing end of the polymer chain and are independent of the type of comonomer at the penultimate position.
6. Reaction rate is independent of chain length.
7. For the chain transfer (except chain transfer to monomers, and comonomers) and chain deactivation parameters, the reactivity of a living polymer chain ended with both monomers and comonomers are the same.

The copolymerization kinetics reaction mechanism is outlined in Table 1, encompassing five types of reactions and a total of

Table 1. Polymerization Reaction Equation

Reaction type	Reaction equation expression	Rate constant
chain activation	$C + Al \xrightarrow{k_a} C^*$	$k_a(n)$
chain initiation	$C^* + M_j \xrightarrow{k_{ij}} P_{1,j}$	$k_{ij}(n)$
chain propagation	$P_{r,j} + M_j \xrightarrow{k_{pij}} P_{r+1,j}^*$	$k_{pij}(n)$
chain transfer	$P_{r,j} + M_j \xrightarrow{k_{M,j}} P_{1,j} + D_{r,j}$	$k_{tM,j}(n)$
	$P_{r,j} + H_2 \xrightarrow{k_{H,j}} P_{0,j} + D_{r,j}$	$k_{tH,j}(n)$
	$P_{r,j} + Al \xrightarrow{k_{Al,j}} P_{0,j} + D_{r,j}$	$k_{tAl,j}(n)$
	$P_{r,j} \xrightarrow{k_{t,j}} P_{0,j} + D_{r,j}$	$k_{t\beta,j}(n)$
chain deactivation	$P_{r,j} \xrightarrow{k_{d,j}} C_d + D_r$	$k_{d,j}(n)$

19 independent reactions: chain activation, chain initiation, chain propagation, chain transfer, and chain deactivation. The chain activation represents the activation of the catalyst by the cocatalyst, while the chain initiation represents the first insertion of the monomer/comonomer into the activated catalyst, leading to an active polymer segment. The chain propagation means the following insertion of a monomer/comonomer into the active polymer segment. The chain transfer is the reaction from an active polymer segment to a deactivated polymer segment and an active catalyst site. It includes chain transfer to monomers, comonomers, hydrogen, cocatalyst, and β -H elimination; they will determine the microstructure of the polymer chain and affect the polymer properties. The chain deactivation refers to the deactivation of active polymer segments, which means that the monomer or comonomer cannot be further inserted into the polymer segments. In Table 1, M_j is the monomer ($j = A$) and comonomer ($j = B$). $P_{r,j}$ and $D_{r,j}$ are the active segments and the deactivated segment respectively, where r is the chain length and j is the end group. Al is the cocatalyst, C is the catalyst, C^* is the active site, and C_d is the deactivated site.

The model is developed using a Python-based open-source optimization modeling language,²³ known for its effectiveness in crafting sophisticated optimization and analysis tools. The constructed polymerization reaction kinetic model integrates population balance model,²⁴ method of moments,²⁵ instantaneous distribution method,⁴ Arrhenius equation, and phase equilibrium database. The method of moments is used to calculate the average properties of a polymer. The i_{th} order moments of live polymer chains and dead polymer chains are defined as eq 2. The instantaneous distribution method is based on the assumption that the polymerization reaction time is much longer than the life span of the polymer chain. The Flory distribution is commonly used to describe the instantaneous chain length distribution in the olefin polymerization process as shown in eq 3.

$$P_A^i = \sum_{r=1}^{\infty} r^i [P_{r,A}]; P_B^i = \sum_{r=1}^{\infty} r^i [P_{r,B}]; D^i = \sum_{r=1}^{\infty} r^i [D_r] \quad (2)$$

$$w(r) = r\tau_j^2 \exp(-r\tau_j) \quad (3)$$

The developed model elucidates the batch dynamic of copolymerizing ethylene and α -olefins, considering the temperature fluctuations in the polymerization reactor and the consequent variations in the gas–liquid equilibrium of ethylene due to temperature changes. Mass balance equations for each component and the moment of each order are outlined in Table S1 of the Supporting Information, with detailed nomenclature available in the Supporting Information. As described above, the constructed model is a DAE model to be tackled by the *Simulator* module utilizing the *IDAS* solver in the *CasADi* framework.

After the simulation of the DAE model, it will yield several polymer property parameters, including number average molecular weight (M_n), weight average molecular weight (M_w), *PDI*, and *MWD*. The monomer and comonomer consumption rate (R_{pA} and R_{pB}) and X_B , F_B can also be obtained. *MWD* is determined based on the Flory distribution equation. For batch or semibatch polymerization processes, Flory distribution only works instantaneously.⁴ It is assumed that the instantaneous MWD_t follows the Flory distribution. The overall *MWD* is derived by accumulating instantaneous MWD_t . The corresponding equations can be found in Table 2.

2.2. Methodology for Parameter Estimation. The kinetic model includes five types of reactions with a total of

19 independent reactions. Given the interdependence of each reaction and their distinct impact on various targets, simultaneous estimation of all kinetic parameters proves to be challenging. It is because constructing a suitable evaluation function that can reflect the influences of all kinetic parameters is a challenging task. To enhance computational efficiency and precision, it becomes imperative to devise a tailored parameter estimation strategy. This paper employs a strategy of categorizing all estimated parameters into multiple subsets, where each subset corresponds to specific targets.¹ Corresponding evaluation functions are designed for step-by-step estimation. Table 3 outlines the kinetic parameters influencing each specific production target.

Table 3. Main kinetic parameters related to different production targets

Production target	Kinetic parameters
Production rate	Chain activation, chain propagation, chain deactivation constants
Number-average molecular weight	Chain transfer constant—to monomer and hydrogen
Monomer conversion rate	Monomer chain propagation constant
Polymer particle density comonomer	Chain propagation constant
Polydispersity index	Chain transfer constant

2.2.1. Strategy of Estimation Parameter. Figure 2 illustrates the developed methodology for estimating kinetic

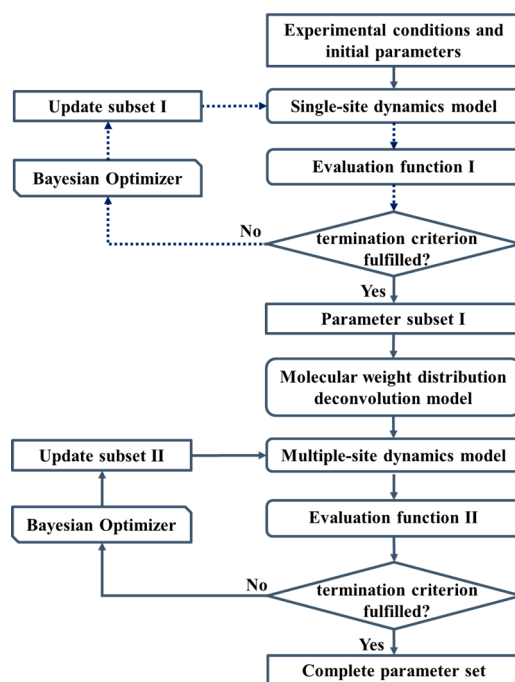


Figure 2. Parameter estimation framework.

parameters from semibatch experimental data. The kinetic parameter set is divided into two parameter subsets, and different evaluation functions are adopted respectively. In the first step, the optimization method introduced in Figure 1 is applied in a single-site kinetic model to estimate the subset I, represented by the dashed line. Once the evaluation function I result meets the convergence criterion or the number of iterations reaches the maximum, it proceeds to the second

Table 2. Calculation Results of Dynamic Model

Object	Expression
R_{pA}	$(k_{iA}[C^*] + k_{pAA}P_A^0 + k_{pBA}P_B^0)[M_A]$
R_{pB}	$(k_{iB}[C^*] + k_{pAB}P_A^0 + k_{pBB}P_B^0)[M_B]$
X_B	$\int R_{pB} dt / [M_B]_0$
$\bar{F}_B$	$\int R_{pB} dt / (\int R_{pB} dt + \int R_{pA} dt)$
$\bar{P}_w$	$(P_A^2 + P_B^2 + D^2) / (P_A^1 + P_B^1 + D^1)$
$\bar{P}_n$	$(P_A^1 + P_B^1 + D^1) / (P_A^0 + P_B^0 + D^0)$
M_n	$\bar{P}_n[m_A(1 - \bar{F}_B) + m_B\bar{F}_B]$
M_w	$\bar{P}_w[m_A(1 - \bar{F}_B) + m_B\bar{F}_B]$
<i>PDI</i>	$(P_A^2 + P_B^2 + D^2)(P_A^0 + P_B^0 + D^0) / (P_A^1 + P_B^1 + D^1)^2$
<i>MWD</i>	$\frac{\int_t^{t+\Delta t} w \log MWD(t) R_p(t) dt}{\int_t^{t+\Delta t} R_p(t) dt}$

step. The MWD deconvolution model is used to obtain the number of catalyst active sites, the mass fraction of each site, and the τ_i . Then a multiple-site kinetic model is obtained. The parameters in the subset I for each site can be obtained by eq 4:¹

$$k_i = km_i \text{SITES} \quad (4)$$

where m_i is the mass fraction for each site, SITES is the number of catalyst active sites, and k is the known single-site kinetic parameters. Finally, the BO algorithm is also applied to estimate the kinetic parameter subset II, using the multisite kinetic model and the evaluation function II. The majority of parameters in the parameter subset II are calculated by eq 4, yet β -H elimination rate constants are estimated separately for each site.

To expedite solutions and enhance feasibility, sensitivity analysis and reasonable assumptions are employed to screen the parameters requiring estimation. Upper and lower bounds for the kinetic parameters are set based on the public data in the reference literature and the preliminary debugging results,^{26–30} effectively constraining the parameter space within a reasonable range.

2.2.2. Selection of Evaluation Function. In the reaction of olefin copolymerization, the consumption rate of monomers and comonomers is predominantly governed by the rate of chain propagation and chain deactivation rate, and the impact of the chain transfer reaction is negligible. The ratio of the monomer chain propagation to the comonomer chain propagation in copolymerization reactions determines important production indicators such as X_B and F_B . In addition, the rate of chain transfer reactions significantly impacts MWD, PDI, and M_n .

Based on the preceding analysis, the kinetic parameters of chain initiation, chain propagation, and chain deactivation parameters are categorized as parameter subset I. Note that the kinetic parameters for each reaction include both the pre-exponential factor and the corresponding activation energy. The kinetic parameters of chain transfer are parameter subset II, including chain transfer to monomer, comonomer, cocatalyst, hydrogen, and β -H elimination. For the parameter subset I, the evaluation function I simultaneously considers the ethylene consumption curve, comonomer conversion rate, and comonomer content of the polymer as evaluation criteria. It is obtained by summing up the mean square error (MSE) between the experimental value and the calculated value via the kinetic model as shown by eq 5.

$$\begin{aligned} \text{EVA1} = \frac{1}{N} \sum_{n=1}^N \left\{ \frac{a}{M} \sum_{m=1}^M [R_{pA,n,m}^{\text{cal}} - R_{pA,n,m}^{\text{exp}}]^2 \right. \\ \left. + b[X_{B,n}^{\text{cal}} - X_{B,n}^{\text{exp}}]^2 + c[F_{B,n}^{\text{cal}} - F_{B,n}^{\text{exp}}]^2 \right\} \quad (5) \end{aligned}$$

where a , b , and c are weight factors, N denotes the experiment set ($n \in N$), and M represents the amount of data ($m \in M$) for the n_{th} experiment. The weighting factors are adjusted to ensure that each component maintains the same order of magnitude. Evaluation function II uses the MSE between the experimentally measured GPC curve and the MWD curve calculated by the model, as shown in eq 6. The relevant information on evaluation functions I and II is outlined in Table 4.

$$\text{EVA2} = \frac{1}{N} \sum_{n=1}^N \left\{ \frac{1}{M} \sum_{m=1}^M [\text{MWD}_{n,m}^{\text{cal}} - \text{MWD}_{n,m}^{\text{exp}}]^2 \right\} \quad (6)$$

Table 4. Parameter Subsets Corresponding to the Evaluation Function and Production Targets

Category	Parameter Subset	Content
Evaluation function I	Chain initiation, Chain propagation, Chain deactivation	R_{pA} , X_B , F_B
Evaluation function II	Chain transfer	MWD

2.3. Bayesian Optimization Algorithm and Application. Bayesian optimization algorithm is an effective global optimization algorithm. It constructs a probabilistic model that represents the surrogate model between the decision variables and the objective function. The optimization operates by iteratively incorporating sample points to correct the surrogate model. Based on this model, the subsequent sampling point for evaluation is obtained by maximizing the acquisition function.

Currently, the most commonly used surrogate model is the Gaussian process (GP),³¹ with acquisition functions include Upper Confidence Bound (UCB),³² Expected Improvement (EI),³³ and Probability of Improvement (PI).³⁴ In this article, the surrogate model employs the GP, and the acquisition function utilizes the UCB function. The Gaussian process, a prevalent non-parametric model, comprises a mean function and a covariance function. Through regression with numerous sampling points, a surrogate function that approximates the black box model is constructed. In the process of selecting the subsequent evaluation point, the UCB function weighs the position of the next optimal evaluation point (i.e., the high mean $u(\lambda)$ interval and the high variance $\sigma(\lambda)$ interval). By setting an additional hyperparameter $Kappa$ to balance utilization and exploration behaviors, as shown by eq 7.

$$\text{UCB}(\lambda) = u(\lambda) + Kappa \cdot \sigma(\lambda) \quad (7)$$

In the developed framework, the BO algorithm is employed to find the optimal kinetic parameters by minimizing the error between the calculated values via the dynamic model and the experimental results. During the optimization process, a Gaussian process model, which is much simpler than the kinetic model, represents the relationship between the decision variables (the kinetic parameters) and the objective function (the error). The accuracy of the surrogate model in terms of representing the characteristics of the kinetic model increases with more evaluated results of the objective function and decision variables. Furthermore, the selection of the decision variables for evaluation is not random. It is determined based on the current surrogate model and the maximization process of the acquisition function, which considers the trade-off between exploration and exploitation. This method circumvents the challenges associated with solving the original complex nonlinear optimization problem and ensures that the results are calculated based on strict mechanisms.

3. RESULTS AND DISCUSSION

3.1. Case Study. The parameter estimation framework proposed in this article is based on the BO algorithm, employing distinct evaluation functions for estimating different parameter subsets. Specifically, evaluation function I mainly determines the production rate, yield, and product grades,

while evaluation function II focuses on the microscopic properties of the polymer. In the following sections, three cases are used to show the effectiveness of the proposed framework with both noise-free data and experimental data. Case 1 involves a simplification of the complete parameter estimation framework described in Section 2.2, using only one evaluation function, to estimate chain activation, chain propagation, and chain deactivation parameters. Case 2 is executed based on the complete parameter estimation framework, fitting experimental data with different batch polymerization reaction times. Case 3 uses experimental results derived from various polymerization reaction temperatures.

It is worth noting that metallocene catalysts are commonly considered as single-site catalysts. However, it has been reported that in supported metallocene catalysts, the polymer exhibits a broad MWD due to the influence of mass and heat transfer within the catalyst.^{35,36} Even in a homogeneous reaction, the product may exhibit multiple active sites due to the interaction between the catalyst precursor and cocatalyst components.^{37–40} Mehdiabadi⁴¹ found that when Al/Ti is lower than 20 000, the PDI tends to increase with decreasing Al/Ti ratio. At very low Al/Ti ratios, the metallocene catalyst loses its single-site characteristics. Thus, when fitting experimental results involving metallocene catalysts, a multisite model might be required to accurately depict the results of the polymerization process.

3.1.1. Case 1. In Case 1, evaluation function I is adopted, which is the sum of the MSE between the experimental values and simulated values of the R_{pA} , X_B , F_B . The upper and lower bounds of the estimated kinetic parameters and fixed value for other parameters are outlined in Table 5, where r_A and r_B represent the reactivity ratio of ethylene and α -olefins, and E_1 , E_2 , and E_3 represent the activation energy constant of chain propagation, chain transfer, and chain deactivation, respectively. Note that for kinetic parameters without lower and upper bounds, the value is fixed. For the BO process, 200 sets of random initial values are selected, and the optimization is iteratively performed 500 times. The optimized ethylene consumption curve results are shown in Figure 3. Notably, the fitting curve closely aligns with the experimental curve, showcasing a high degree of overlap. However, since the kinetic model may not entirely capture the intricacies of the real experimental process and the solution time is limited, a certain deviation between the two curves persists. Nevertheless, the case demonstrates the capability of the proposed method to effectively obtain kinetic parameters for the essential characteristics of olefin polymerization processes.

3.1.2. Case 2. Case 2 employs a complete parameter estimation framework, as depicted in Figure 2. The kinetic parameter subset related to evaluation functions I and II is shown in Table 6. The corresponding lower and upper bounds and fixed values are shown in Table S2 of the Supporting Information. Three batch experiments with different polymerization reaction times (reaction times of 300, 500, and 700 s, respectively) are considered, namely, Exp-TIME-300, Exp-TIME-500, and Exp-TIME-700. Once the value of parameter subset I is determined through the single-site kinetic model, the MWD deconvolution model is used to deconvolve the GPC curve to obtain the number of active sites, $ther_i$, of each site, and the mass fraction of each site, the data is outlined in Table S3 of the Supporting Information. Then a multi-active site model is established, using MWD as the evaluation criterion to estimate the parameter subset II and obtain the

Table 5. Lower and Upper Bounds and Fixed Value of Kinetic Parameters for Case 1

Parameter	k_a	k_{iA}	k_{iB}	k_{pAA}	k_{pBB}	r_A	r_B	k_{tAA}	k_{tAB}	k_{tBA}	k_{tBB}	k_{tH}	k_{tP}	k_{tAl}	k_d	E_1	E_2	E_3
Lower bound	—	0	—	500	0	1	0.05	—	—	—	—	—	—	—	0.0001	10000	—	10000
Upper bound	—	10	—	1500	500	60	1	—	—	—	—	—	—	—	0.01	100000	—	100000
Fixed value	10000	—	0	—	—	—	—	0.001	0.001	0.001	0.001	0	1	0.0018	—	—	58600	—

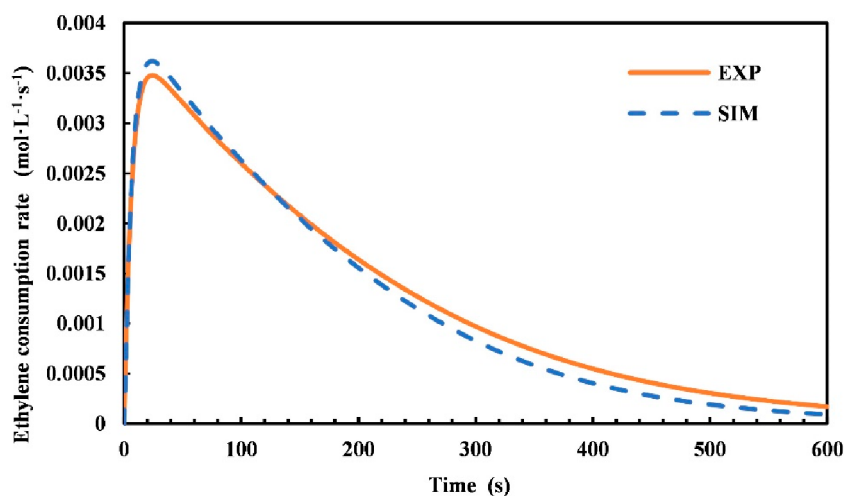


Figure 3. Case 1: Fitting results of experimental and simulated values of the ethylene consumption curve.

Table 6. Case 2: Parameter Subset and Related Parameters

Category	Parameter							
Parameter subset 1	k_{iA}	k_{pAA}	k_{pBB}	r_A	r_B	k_d	E_1	E_3
Parameter subset 2	k_{iAA}	k_{iAB}	k_{iBA}	k_{iBB}	$k_{i\beta(1)}$	$k_{i\beta(2)}$	$k_{i\beta(3)}$	E_2

complete set of parameters. The parameter subset II includes chain transfer to monomers, comonomers, hydrogen, cocatalyst, and β -H elimination and the corresponding activation energy, as shown in Table 6. The fitting results of the MWD data derived from Exp-TIME-500 are shown in Figure 4, and the fitting results of ethylene consumption curves for Exp-TIME-300, Exp-TIME-500, and Exp-TIME-700 are shown in Figure 5. The experiment and simulated results are represented by the solid and dashed lines, respectively. The same representation method is adopted in the following cases.

In Case 2, the kinetic parameters are fitted using three experiments with different reaction time. Three reaction kinetic models under different experimental conditions are established correspondingly, with the same kinetic parameters. The values of the obtained kinetic parameters and activation energy parameters are listed in Table S4 of the Supporting

Information. Figure 5 shows a deviation caused by excessive ethylene injection initially due to the stamping process. This led to higher reactor pressure, disrupting ethylene flow and resulting in a fluctuating curve. Although our model does not account for stamping, we have attempted to capture the characteristics of the curve, even though they cannot align precisely with the real curve.

3.1.3. Case 3. Kinetic parameter estimation using three experiments (namely, Exp-TEMP-125, Exp-TEMP-145, and Exp-TEMP-155) under different temperatures (125, 145, and 155 °C) is performed, where the complete framework of Figure 2 is applied. The fitting results of the ethylene consumption curve for Exp-TEMP-125, Exp-TEMP-145, and Exp-TEMP-155 and MWD for Exp-Temp-125 are shown in Figures 6 and 7, and the deconvolution data are outlined in Table S5 of the Supporting Information. The lower and upper bounds and the obtained values of the kinetic parameters are listed in Tables S5 and S6 of Supporting Information, respectively. It is found that a relatively large deviation exists in the initial stage of the experiments, as shown in Figure 6. The reason is when the initial pressure of the reactor is smaller than the set pressure of the pressure controller, it will lead to a brief stamping pressure. The ethylene consumption curve is

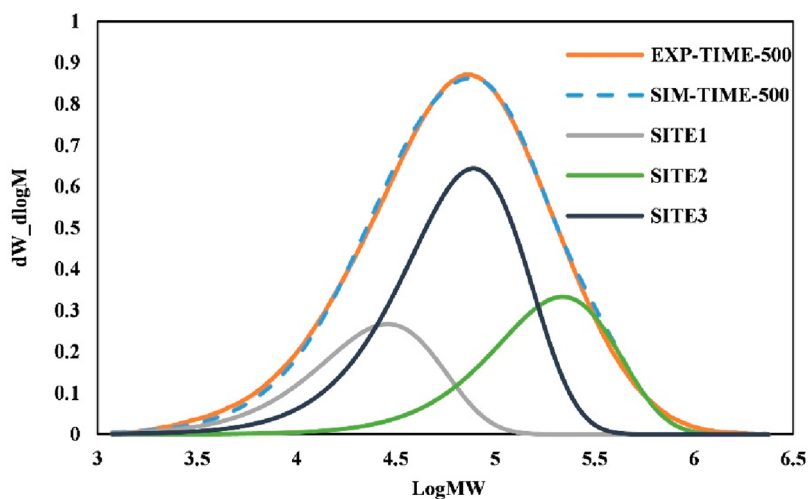


Figure 4. Case 2: MWD fitting results.

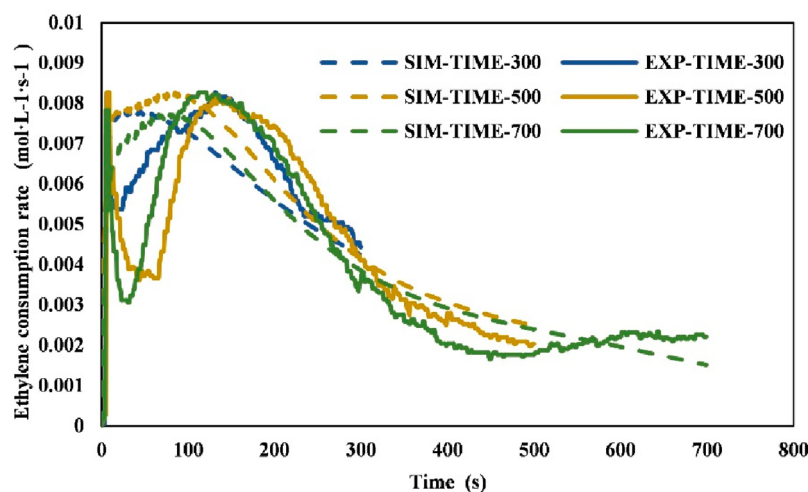


Figure 5. Case 2: Fitting results of the ethylene consumption curve.

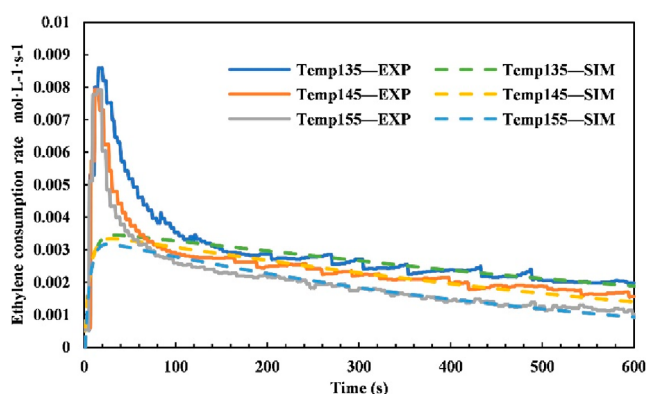


Figure 6. Case 3: Fitting results of the ethylene consumption curve.

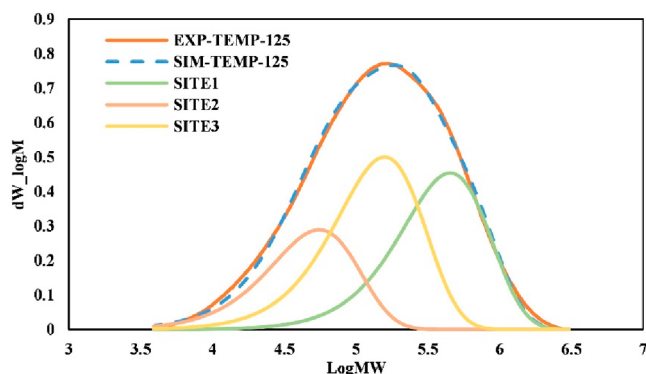


Figure 7. Case 3: MWD fitting results.

generally regarded as the ethylene flow curve into the reactor. In this scenario, the ethylene flow rate may momentarily surge in the preliminary stage of the reaction and then decline, but it does not represent the actual reaction process. Since the kinetic polymerization model assumes no gas–liquid mass transfer resistance and does not include the initial pressurization process, the obtained ethylene consumption curve cannot replicate such a phenomenon. In addition to the initial stage, the obtained ethylene consumption curves for three experiments under reaction temperatures of 135, 145, and 155 °C fit quite well with the experiment results.

3.2. Different Parameter Estimation Strategies and Extrapolation Results. To demonstrate the advantages of the proposed approach, a comparative analysis of parameter estimation strategies between the proposed approach and alternative methods is presented. Case 2 and Case 3 from previous sections are used for the comparison. For Case 2, four cases (Cases 2-C1, C2, C3A, and C3B) are used for comparisons. In Table 7, the strategy I represents the fitting methods of using regression equations to estimate the chain deactivation parameter in literatures.^{16,17,26,35,42} The specific content is introduced in Section 5 of the Supporting Information. The strategy II indicates that the kinetic model considers the variations of the temperature and ethylene concentration during the experimental process. Compared to Case 2, Case 2-C1 does not adopt the strategy II, Case 2-C2 adopts the strategy I, and Case 2-C3A and C3B use Baron rather than the developed BO method. Similarly, Case 3-C1 and Case 3-C2 are used to compare with Case 3.

Table 7. Comparison of Different Parameter Estimation Strategies

Case name	Strategy I	Strategy II	Solver	Fitting result (EVA1)	Calculation time and iterations	Extrapolation result 1 (EVA1)	Extrapolation result 2 (EVA1)
Case2	No	Yes	BO	19.98	2477 s/500	4.25	6.43
Case2-C1	No	No	BO	21.66	1512 s/500	5.66	6.95
Case2-C2	Yes	Yes	BO	20.50	2359 s/500	4.53	6.62
Case2-C3A	No	Yes	Baron	32.13	2477 s	8.46	10.82
Case2-C3B	No	Yes	Baron	25.74	7200 s	6.36	8.52
Case3	No	Yes	BO	4.65	2455 s/500	1.51	1.05
Case3-C1	No	No	BO	5.22	1682 s/500	1.83	1.22
Case3-C2	Yes	Yes	BO	4.89	2385 s/500	1.72	1.13

The results in Table 7 show that cases (Cases 2-C1 and Case 3-C1) without considering temperature variation in the dynamic model (strategies II) have higher errors than Cases 2 and 3, respectively. When the strategy I is applied in Case 2-C2 and Case 3-C2, the fitting results are also worse than the ones by Case 2 and Case 3, respectively, shown by higher evaluation function values. The above comparisons are carried out using the same iteration number of BO algorithms, and the calculation time is also shown in Table 7. Furthermore, in Case 2-C3A and C3B, a commercial deterministic algorithm (Baron) is applied with 2477 and 7200 s as the maximum running time. The results indicate that the kinetic parameter estimation problem cannot be efficiently solved with the same calculation time and even three times of calculation time.

To verify the extrapolation effectiveness of the obtained kinetic parameters, two experimental results, which are not used in the kinetic parameter estimation process, are tested through the models obtained in different cases in Table 7. The extrapolation results using evaluation function I as a criterion (eq 5) are summarized in Table 7. Extrapolation result 1 and extrapolation result 2 represent the EVA1 result of Exp-TIME-1200 and Exp-TIME-1800 for Case 2 (Polymerization experiments with reaction times of 1200 s and 1800 s) and Exp-TEMP-125 and Exp-TEMP-160 for Case 3 (Polymerization experiments with reaction temperature of 125 and 160 °C). It shows that the BO algorithm outperforms the commercial deterministic algorithm within equivalent computational times, as the MSE of Case2–3 is much higher than that of Case 2. Additionally, the proposed parameter estimation strategy in Case 2 and Case 3 is better than the ones in Cases 2-C1 to C2 and Cases 3-C1 to C2, respectively, demonstrated by lower MSE values. Two evaluation criteria for extrapolation results also indicate that the obtained model via Case 2 and Case 3 has better extrapolation accuracy.

The information contained in MWD properties surpasses that of general scalar indicators (e.g., molecular weight), making MWD a preferred objective function. For the estimation of kinetic parameters for chain transfer, two cases (Cases 2-C5 and C6) using PDI and M_n for each active site as the indicator in the evaluation function II are compared with Case 2, which uses MWD (eq 6). Table 8 displays the EVA2

Table 8. Comparison of Different Evaluation Functions

Case name	Indicator in evaluation function II	Fitting result (EVA2)	Extrapolation result 1 (EVA2)	Extrapolation result 2 (EVA2)
Case2	MWD	0.0845	0.144	0.629
Case2-C5	M_n	0.991	1.143	1.183
Case2-C6	PDI	2.155	2.986	3.166

results for Case 2, Case 2-C5, and Case 2-C6. Notably, the utilization of MWD in Case 2 lead to better results with smaller MSE of MWD. Figure 8 illustrates the extrapolation results of EXP-TIME-300 and EXP-TIME-1200 using kinetic parameters obtained via Case 2, and the EVA2 (MSE of MWD) calculated through eq 6 is 0.144 and 0.629, respectively. Overall, the parameter estimation strategy proposed in this work performs better in both fitting results and extrapolating results. Using PDI or M_n as the objective function might result in multiple parameter combinations. The same PDI or M_n could correspond to various MWD. This

complexity poses challenges in accurate fitting and complicates the extrapolation effect of the kinetic model. Consequently, relying on PDI or M_n might hinder the precision of the fit and compromise the predictive capability of the kinetic model.

3.3. Algorithm Comparisons. The dynamic polymerization kinetic model considering the temperature variation is important for the accuracy of the kinetic parameter estimation, as demonstrated by the comparisons in Section 3.2. The Bayesian optimization algorithm is also a critical factor in the developed framework in Figure 1. In this section, the BO algorithm is compared with two deterministic algorithms (Sequential least-squares programming algorithm/SLSQP and Interior Point Optimizer/IPOPT) and two stochastic algorithms (Particle Swarm Optimization algorithm/PSO, and Genetic algorithm/GA).

To conduct comparisons, Case 2 from the previous section, including three batch experiments (Exp-TIME-300, Exp-TIME-500, and Exp-TIME-700), are adopted. It is known that the selection of initial values significantly impacts the resolution of Nonlinear Programming (NLP) problems when utilizing SLSQP and IPOPT. Therefore, each algorithm is carried out 20 times using different initial values. The BO, GA and PSO are also carried out 20 times, while the number of iterations is set to 300 and 500. The deterministic algorithm exhibits varying numbers of iterations for each calculation with a cumulative total of 4400 iterations. Table 9 presents both the average and the best evaluation function values achieved along with the average iteration count and the iteration count corresponding to the best outcome when employing SLSQP, BO, GA, and PSO.

While the best outcome achieved by SLSQP surpasses that of BO, it is noteworthy that SLSQP relies on the selection of initial values. The average result of 20 calculations is much higher than the one of BO. The optimal solution is obtained on the 11th calculation, and it has been iterated 2282 times by that point. Additionally, no feasible solutions can be found within the set maximum number of gradient calculations when IPOPT is used. When comparing the BO algorithm with GA and PSO algorithms, although the results of BO and GA are almost the same after 500 iterations, the efficiency of BO is better at 300 iterations, which means that the BO algorithm is more suitable in fast prediction scenarios.

4. CONCLUSION

This work proposes a dynamic polymerization model and a framework for estimating kinetic parameters. The model considers the non-steady-state olefin polymerization process, incorporating temperature fluctuations. Employing the Bayesian optimization algorithm within the parameter estimation framework enables the identification of optimal kinetic parameters by minimizing the MSE between the obtained results of the dynamic model and the experimental results. Specifically, the ethylene flow curve, comonomer conversion rate, and comonomer content of the polymer product are used in evaluation function I for the kinetic parameter estimation of chain initiation, chain propagation, and chain deactivation constants. Besides, MWD is considered in evaluation function II for the estimation of chain transfer parameters. The framework's effectiveness and advantages are demonstrated by fitting multiple sets of noise-free data and experimental values. The proposed method exhibits better fitting and extrapolation effects than those of several reported strategies and algorithms. Utilizing the Bayesian optimization algorithm,

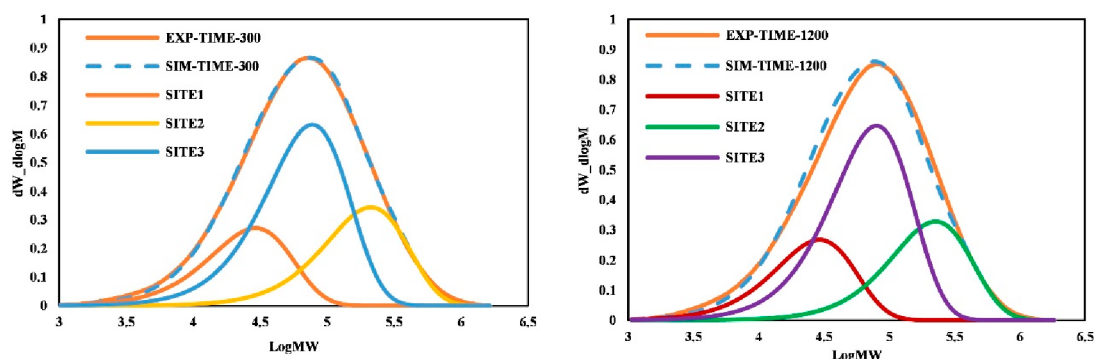


Figure 8. Extrapolation results of EXP-TIME-300 and EXP-TIME-1200 using kinetic parameters obtained via Case 2.

Table 9. Comparisons of Results Using Various Algorithms for Case 2

Algorithm	Fitting results (Average)	Fitting results (Best)	Iterations
BO (this paper)	21.22	19.98	300
BO (this paper)	20.18	19.98	500
SLSQP	477.30	17.13	220 ^a ; 378 ^b
IPOPT			30000 ^c
GA	23.16	20.64	300
GA	20.33	19.98	500
PSO	32.56	30.10	300
PSO	25.61	24.61	500

^aAverage number of iterations for 20 runs. ^bThe number of iterations used in the calculation to obtain the optimal solution. ^cNumber of gradient calculations.

the proposed approach sequentially estimates multiple subsets of kinetic parameters by employing diverse evaluation functions. The methodology not only enhances computational efficiency but also improves extrapolation capabilities.

In future work, we aim to extend the kinetic model to consider the mass transfer resistance, which can reflect the gas–liquid equilibrium more accurately. Besides, to achieve the goal of the application in an automated high-throughput experiment for expediting research and development, the efficiency of the proposed method should be further enhanced and improved.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.3c04665>.

Mass balance equations for each component and the moment of each order; calculation equation of kinetic parameters k ; EXP-TIME related data; EXP-TEMP related data; nomenclature; method of regression equations to estimate kinetic parameters (PDF)

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Notes

The authors declare no competing financial interest.

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